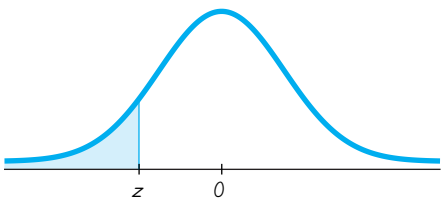


# NEGATIVE z Scores



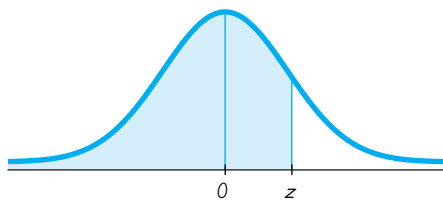
**TABLE A-2** Standard Normal (z) Distribution: Cumulative Area from the LEFT

| z               | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| −3.50 and lower | .0001 |       |       |       |       |       |       |       |       |       |
| −3.4            | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0002 |
| −3.3            | .0005 | .0005 | .0005 | .0004 | .0004 | .0004 | .0004 | .0004 | .0004 | .0003 |
| −3.2            | .0007 | .0007 | .0006 | .0006 | .0006 | .0006 | .0006 | .0005 | .0005 | .0005 |
| −3.1            | .0010 | .0009 | .0009 | .0009 | .0008 | .0008 | .0008 | .0008 | .0007 | .0007 |
| −3.0            | .0013 | .0013 | .0013 | .0012 | .0012 | .0011 | .0011 | .0011 | .0010 | .0010 |
| −2.9            | .0019 | .0018 | .0018 | .0017 | .0016 | .0016 | .0015 | .0015 | .0014 | .0014 |
| −2.8            | .0026 | .0025 | .0024 | .0023 | .0023 | .0022 | .0021 | .0021 | .0020 | .0019 |
| −2.7            | .0035 | .0034 | .0033 | .0032 | .0031 | .0030 | .0029 | .0028 | .0027 | .0026 |
| −2.6            | .0047 | .0045 | .0044 | .0043 | .0041 | .0040 | .0039 | .0038 | .0037 | .0036 |
| −2.5            | .0062 | .0060 | .0059 | .0057 | .0055 | .0054 | .0052 | .0051 | *     | .0049 |
| −2.4            | .0082 | .0080 | .0078 | .0075 | .0073 | .0071 | .0069 | .0068 | ↑     | .0066 |
| −2.3            | .0107 | .0104 | .0102 | .0099 | .0096 | .0094 | .0091 | .0089 |       | .0087 |
| −2.2            | .0139 | .0136 | .0132 | .0129 | .0125 | .0122 | .0119 | .0116 |       | .0113 |
| −2.1            | .0179 | .0174 | .0170 | .0166 | .0162 | .0158 | .0154 | .0150 |       | .0146 |
| −2.0            | .0228 | .0222 | .0217 | .0212 | .0207 | .0202 | .0197 | .0192 |       | .0188 |
| −1.9            | .0287 | .0281 | .0274 | .0268 | .0262 | .0256 | .0250 | .0244 |       | .0239 |
| −1.8            | .0359 | .0351 | .0344 | .0336 | .0329 | .0322 | .0314 | .0307 |       | .0301 |
| −1.7            | .0446 | .0436 | .0427 | .0418 | .0409 | .0401 | .0392 | .0384 |       | .0375 |
| −1.6            | .0548 | .0537 | .0526 | .0516 | .0505 | *     | .0495 | .0485 |       | .0475 |
| −1.5            | .0668 | .0655 | .0643 | .0630 | .0618 | ↑     | .0606 | .0594 |       | .0582 |
| −1.4            | .0808 | .0793 | .0778 | .0764 | .0749 |       | .0735 | .0721 |       | .0708 |
| −1.3            | .0968 | .0951 | .0934 | .0918 | .0901 |       | .0885 | .0869 |       | .0853 |
| −1.2            | .1151 | .1131 | .1112 | .1093 | .1075 |       | .1056 | .1038 |       | .1020 |
| −1.1            | .1357 | .1335 | .1314 | .1292 | .1271 |       | .1251 | .1230 |       | .1210 |
| −1.0            | .1587 | .1562 | .1539 | .1515 | .1492 |       | .1469 | .1446 |       | .1423 |
| −0.9            | .1841 | .1814 | .1788 | .1762 | .1736 |       | .1711 | .1685 |       | .1660 |
| −0.8            | .2119 | .2090 | .2061 | .2033 | .2005 |       | .1977 | .1949 |       | .1922 |
| −0.7            | .2420 | .2389 | .2358 | .2327 | .2296 |       | .2266 | .2236 |       | .2206 |
| −0.6            | .2743 | .2709 | .2676 | .2643 | .2611 |       | .2578 | .2546 |       | .2514 |
| −0.5            | .3085 | .3050 | .3015 | .2981 | .2946 |       | .2912 | .2877 |       | .2843 |
| −0.4            | .3446 | .3409 | .3372 | .3336 | .3300 |       | .3264 | .3228 |       | .3192 |
| −0.3            | .3821 | .3783 | .3745 | .3707 | .3669 |       | .3632 | .3594 |       | .3557 |
| −0.2            | .4207 | .4168 | .4129 | .4090 | .4052 |       | .4013 | .3974 |       | .3936 |
| −0.1            | .4602 | .4562 | .4522 | .4483 | .4443 |       | .4404 | .4364 |       | .4325 |
| −0.0            | .5000 | .4960 | .4920 | .4880 | .4840 |       | .4801 | .4761 |       | .4721 |

NOTE: For values of z below −3.49, use 0.0001 for the area.

\*Use these common values that result from interpolation:

| z score | Area     |
|---------|----------|
| −1.645  | 0.0500 ← |
| −2.575  | 0.0050 ← |



# POSITIVE z Scores

**TABLE A-2** (continued) Cumulative Area from the LEFT

| z           | .00   | .01   | .02   | .03   | .04   | .05    | .06   | .07   | .08    | .09   |
|-------------|-------|-------|-------|-------|-------|--------|-------|-------|--------|-------|
| 0.0         | .5000 | .5040 | .5080 | .5120 | .5160 | .5199  | .5239 | .5279 | .5319  | .5359 |
| 0.1         | .5398 | .5438 | .5478 | .5517 | .5557 | .5596  | .5636 | .5675 | .5714  | .5753 |
| 0.2         | .5793 | .5832 | .5871 | .5910 | .5948 | .5987  | .6026 | .6064 | .6103  | .6141 |
| 0.3         | .6179 | .6217 | .6255 | .6293 | .6331 | .6368  | .6406 | .6443 | .6480  | .6517 |
| 0.4         | .6554 | .6591 | .6628 | .6664 | .6700 | .6736  | .6772 | .6808 | .6844  | .6879 |
| 0.5         | .6915 | .6950 | .6985 | .7019 | .7054 | .7088  | .7123 | .7157 | .7190  | .7224 |
| 0.6         | .7257 | .7291 | .7324 | .7357 | .7389 | .7422  | .7454 | .7486 | .7517  | .7549 |
| 0.7         | .7580 | .7611 | .7642 | .7673 | .7704 | .7734  | .7764 | .7794 | .7823  | .7852 |
| 0.8         | .7881 | .7910 | .7939 | .7967 | .7995 | .8023  | .8051 | .8078 | .8106  | .8133 |
| 0.9         | .8159 | .8186 | .8212 | .8238 | .8264 | .8289  | .8315 | .8340 | .8365  | .8389 |
| 1.0         | .8413 | .8438 | .8461 | .8485 | .8508 | .8531  | .8554 | .8577 | .8599  | .8621 |
| 1.1         | .8643 | .8665 | .8686 | .8708 | .8729 | .8749  | .8770 | .8790 | .8810  | .8830 |
| 1.2         | .8849 | .8869 | .8888 | .8907 | .8925 | .8944  | .8962 | .8980 | .8997  | .9015 |
| 1.3         | .9032 | .9049 | .9066 | .9082 | .9099 | .9115  | .9131 | .9147 | .9162  | .9177 |
| 1.4         | .9192 | .9207 | .9222 | .9236 | .9251 | .9265  | .9279 | .9292 | .9306  | .9319 |
| 1.5         | .9332 | .9345 | .9357 | .9370 | .9382 | .9394  | .9406 | .9418 | .9429  | .9441 |
| 1.6         | .9452 | .9463 | .9474 | .9484 | .9495 | *.9505 | .9515 | .9525 | .9535  | .9545 |
| 1.7         | .9554 | .9564 | .9573 | .9582 | .9591 | ↑.9599 | .9608 | .9616 | .9625  | .9633 |
| 1.8         | .9641 | .9649 | .9656 | .9664 | .9671 | ↑.9678 | .9686 | .9693 | .9699  | .9706 |
| 1.9         | .9713 | .9719 | .9726 | .9732 | .9738 | ↑.9744 | .9750 | .9756 | .9761  | .9767 |
| 2.0         | .9772 | .9778 | .9783 | .9788 | .9793 | ↑.9798 | .9803 | .9808 | .9812  | .9817 |
| 2.1         | .9821 | .9826 | .9830 | .9834 | .9838 | ↑.9842 | .9846 | .9850 | .9854  | .9857 |
| 2.2         | .9861 | .9864 | .9868 | .9871 | .9875 | ↑.9878 | .9881 | .9884 | .9887  | .9890 |
| 2.3         | .9893 | .9896 | .9898 | .9901 | .9904 | ↑.9906 | .9909 | .9911 | .9913  | .9916 |
| 2.4         | .9918 | .9920 | .9922 | .9925 | .9927 | ↑.9929 | .9931 | .9932 | .9934  | .9936 |
| 2.5         | .9938 | .9940 | .9941 | .9943 | .9945 | ↑.9946 | .9948 | .9949 | *.9951 | .9952 |
| 2.6         | .9953 | .9955 | .9956 | .9957 | .9959 | ↑.9960 | .9961 | .9962 | ↑.9963 | .9964 |
| 2.7         | .9965 | .9966 | .9967 | .9968 | .9969 | ↑.9970 | .9971 | .9972 | ↑.9973 | .9974 |
| 2.8         | .9974 | .9975 | .9976 | .9977 | .9977 | ↑.9978 | .9979 | .9979 | ↑.9980 | .9981 |
| 2.9         | .9981 | .9982 | .9982 | .9983 | .9984 | ↑.9984 | .9985 | .9985 | ↑.9986 | .9986 |
| 3.0         | .9987 | .9987 | .9987 | .9988 | .9988 | ↑.9989 | .9989 | .9989 | ↑.9990 | .9990 |
| 3.1         | .9990 | .9991 | .9991 | .9991 | .9992 | ↑.9992 | .9992 | .9992 | ↑.9993 | .9993 |
| 3.2         | .9993 | .9993 | .9994 | .9994 | .9994 | ↑.9994 | .9994 | .9995 | ↑.9995 | .9995 |
| 3.3         | .9995 | .9995 | .9995 | .9996 | .9996 | ↑.9996 | .9996 | .9996 | ↑.9996 | .9997 |
| 3.4         | .9997 | .9997 | .9997 | .9997 | .9997 | ↑.9997 | .9997 | .9997 | ↑.9997 | .9998 |
| 3.50 and up | .9999 |       |       |       |       |        |       |       |        |       |

NOTE: For values of z above 3.49, use 0.9999 for the area.

\*Use these common values that result from interpolation:

| z score | Area     |
|---------|----------|
| 1.645   | 0.9500 ← |
| 2.575   | 0.9950 ← |

## Common Critical Values

| Confidence Level | Critical Value |
|------------------|----------------|
| 0.90             | 1.645          |
| 0.95             | 1.96           |
| 0.99             | 2.575          |